

A Multi-Signal ML System

Combining price, language, and news signals to monitor TD Bank's relative performance.

Agenda

SECTION I · OVERVIEW

01 Agenda

SECTION II · PROBLEM

02 Business Opportunity

SECTION III · DATA

03 Data Collection & Preprocessing

SECTION IV · LANGUAGE SIGNALS

04 Method

05 AML Case Study

SECTION V · PREDICTIVE MODEL

06 Setup

07 Performance

08 Explainability

SECTION VI · LIMITATIONS & NEXT STEPS

09 Limitations & Next Steps

SECTION VII · EXECUTIVE SUMMARY

10 What We Built

Business Opportunity

An opportunity to systematically combine multiple signals driving relative stock performance.

01

Current State

- Canadian bank equities are a core pension holding.
- Relative performance monitoring relies on manual review of calls, filings, and news.

02

Current Challenge

- No systematic way to combine price, language, and news into one signal.
- Indicators are read in isolation.
- Large text volume makes it easy to miss meaningful information.

03

Our Solution

- A statistically sound ML system reading price, language, and news together.
- Flags when a bank is diverging from its sector peers.
- Trained on 5 years of historical data.
- Capable of scaling

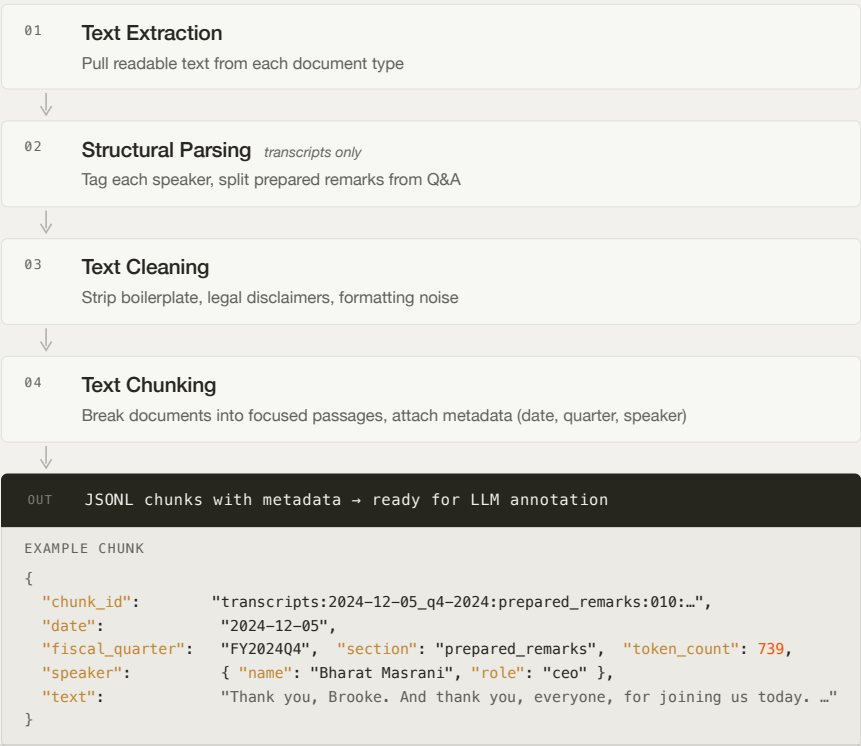
Data Collection & Preprocessing

Data Sources

CONTENT	SOURCE	TIME PERIOD
TD.TO daily price	yfinance API	Feb 2021 → Apr 2026
XFN daily price	yfinance API	Feb 2021 → Apr 2026
Earnings call transcripts (21 calls)	TD Investor Relations	FY2021Q1 → FY2026Q1
Annual filings — Form 40-F (5 filings)	TD Investor Relations	FY2021 → FY2025
Quarterly Reports to Shareholders (16 reports)	TD Investor Relations	FY2021Q1 → FY2026Q1
Press releases (688 articles)	TD Newsroom	Feb 2021 → Apr 2026

Note: TD is a Canadian Foreign Private Issuer. Form 40-F is the annual filing equivalent to a US 10-K; the Quarterly Report to Shareholders is the equivalent to a 10-Q.

Text Preprocessing Pipeline



Language Signals — Method

WHY LLM

- No labeled data needed (zero-shot)
- Understands financial jargon, management framing, and intent — picks up tone that lexical models miss
- Multi-task in one pass — sentiment, topic, and key quote in a single call

MODEL

gpt-4o-mini, fully deterministic

SYSTEM PROMPT

You are a financial analyst specialising in Canadian banks. Analyse the following passage from a TD Bank document and respond with a JSON object. Be concise and precise. Respond ONLY with the JSON — no prose, no markdown fences.

USER MESSAGE (output schema)

Respond with exactly this JSON and nothing else:

```
{
  "sentiment":      "positive" or "neutral" or "negative",
  "sentiment_score": <float -1.0 to 1.0>,
  "topics":         [<1-3 labels>],
  "key_quote":      "<one sentence, max 20 words>"
}
```

SAMPLE RESPONSE

```
{
  "sentiment":      "neutral",
  "sentiment_score": 0.0,
  "topics":         ["regulatory_AML", "Canadian_personal", "macro_outlook"],
  "key_quote":      "We expect to have the majority of the management
remediation actions implemented by end of calendar 2025."
}
```

TOPIC CATEGORIES — 12 labels across 2 groups

- Business performance: margins, credit, capital, segments, costs, M&A
- Forward signals: regulatory/AML, macro outlook, guidance

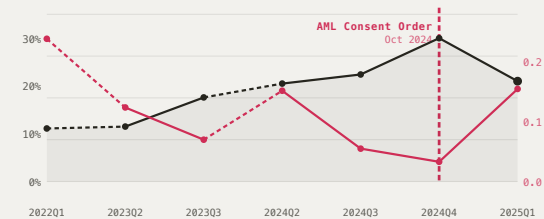
VALIDATION (detail in appendix)

01	Manual sample review
02	Pre-trained financial NLP model comparison
03	Known major event cross-check

Language Signals — AML Case Study

Phase 1 — Build-up FY2022Q1 → FY2025Q1

AML TOPIC SHARE & SENTIMENT SCORE



— AML Share % — AML Sentiment
Solid line = consecutive quarters · Dotted line = trend across skipped quarter(s)
AML discussion doubled. Tone fell 86%. Signal appeared before the headline.

Sources: transcripts + filings + news

+170%

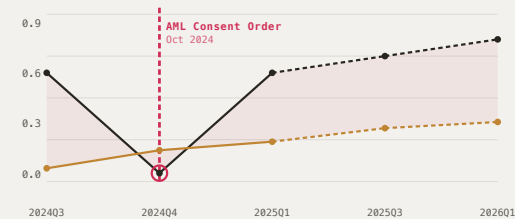
AML TOPIC SHARE — 11.1% (FY2022Q1) TO 30.0% (FY2024Q4)

−86%

AML SENTIMENT — 0.239 (FY2022Q1) TO 0.033 (FY2024Q4)

Phase 2 — Peak Enforcement FY2024Q4

FRAMING GAP — CEO PREPARED REMARKS VS. FILING



— CEO Prep — Filing Tone
Solid line = consecutive quarters · Dotted line = trend across skipped quarter(s)
CEO sentiment hit 0.000 — the only zero in the dataset. Management less optimistic than their own filings.

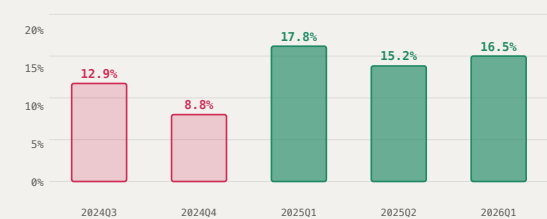
Sources: CEO prepared remarks (transcripts) vs. filings tone

−124%

FRAMING GAP CHANGE AT PEAK — FROM +0.571 (FY2024Q3) TO −0.137 (FY2024Q4)

Phase 3 — Recovery FY2024Q3 → FY2026Q1

GUIDANCE TOPIC SHARE (%)



Framing gap back to +0.494. Guidance at five-year high. Analyst Q&A sentiment reached dataset high of 0.529.

Sources: transcripts + filings + news

+102%

GUIDANCE SHARE INCREASE — 8.8% (FY2024Q4) TO 17.8% (FY2025Q1)

Section V · Predictive Model

Predictive Model — Setup

TARGET

target = TD_5d_return - XFN_5d_return

5-day return = adjusted close at t+5 / adjusted close at t - 1. Adjusted close accounts for dividends and splits.

Subtracting XFN turns the question into: did TD outperform or underperform its Canadian financial-sector benchmark?

XFN is TD's most direct liquid peer benchmark — major Canadian banks are the top holdings.

CLASS LABELS +/-0.3% band

+1 Outperform

TD 5-day excess return > +0.3%

0 Neutral

Within +/-0.3%

-1 Underperform

TD 5-day excess return < -0.3%

Roughly 30–35% of days are neutral, filtering noise while preserving directional moves.

SAMPLE OUTPUT

[0.50, 0.30, 0.20]

P(+1) = 0.50 · P(0) = 0.30 · P(-1) = 0.20

Each model outputs a probability distribution across the three classes that sums to 1. The highest-probability class is selected.

ALGORITHM & FEATURE CHUNKS

XGBoost multiclass

Price Model

6 features

DAILY MARKET BEHAVIOR

- TD trailing returns: 5d, 20d
- TD-vs-XFN relative returns: 5d, 20d
- 52-week high distance
- 20-day volatility

Transcript Model

10 features

EARNINGS CALL & FILLINGS SIGNAL

- Executive tone
- Analyst Q&A sentiment
- Framing gap
- Guidance, AML, and credit-quality topic signals
- Days since call

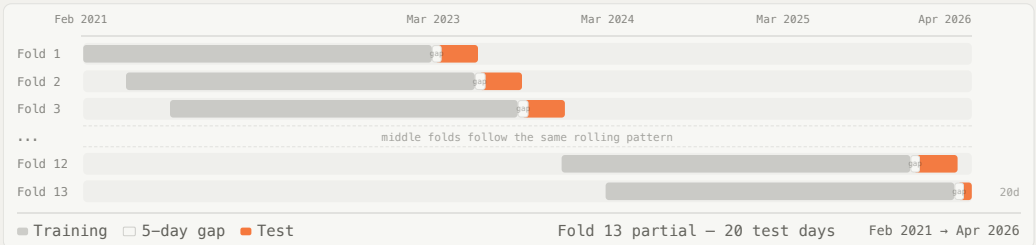
News Model

5 features

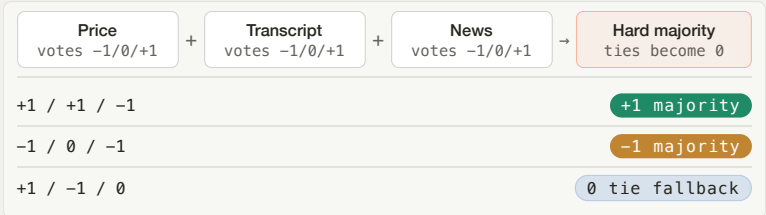
TD PRESS-RELEASE FLOW

- Mean sentiment: 7d, 30d
- Article count: 7d, 30d
- Days since last news

TRAINING STRATEGY Rolling walk-forward · 2-year training window · 5-day gap · 1-quarter test



ENSEMBLE METHODOLOGY



Predictive Model — Performance

Performance view from the model card and xgb_ensemble_v2_performance.ipynb, focused on directional accuracy and baseline lift.

56.6%

DIRECTIONAL ACCURACY

+6.4pp

VS MAJORITY CLASS

+5.5pp

VS MOMENTUM

Aggregate Summary

The highlighted column is the production ensemble. Baseline DA is available for the ensemble comparison, not separately for each sub-model.

Best aggregate DA: 56.6%

	Price	Transcript	News	Hard Majority Vote	Majority Class	Momentum
Directional accuracy Correct sign when a direction call is made; ensemble 95% CI is 53.1%–60.1%	53.0%	55.0%	53.6%	56.6% [53.1%, 60.1%]	50.3% Ensemble lift: +6.4pp	51.1% Ensemble lift: +5.5pp
Directional call rate How often the model makes a +1 or –1 call	97.2%	99.3%	99.6%	98.2%	n/a	n/a

Baseline Context

Majority Class

Predicts the most frequent class from each fold's training window on every test day. This checks whether the ensemble adds signal beyond class-frequency behavior.

Momentum

Continues the sign of the prior 5-day TD-vs-XFN relative return. This is the simple rule-based alternative for a relative-return signal.

Predictive Model — Explainability

What Each Model Is Reading

Price Model

6 features

Distance from 52-week high → far from peak → lean underperformance

20-day return → sustained weakness → lean underperformance

20-day volatility → erratic price action → risk alarm

Transcript Model

10 features

Days since call → long gap — info decays → push toward neutral

Analyst Q&A tone → constructive session → lean outperformance

Framing gap → CEO more upbeat than filings → lean outperformance

News Model

5 features

30-day article volume → coverage burst → lean underperformance

30-day sentiment → sustained positive tone → lean outperformance

Where the Model Holds Up and Where It Breaks

When It Works

Best regime: clear, stable conditions

- TD-specific behavior is easier to isolate from sector noise
- Volatility is orderly, not shock-driven
- Signals across price, transcript, and news are coherent
- The model has seen patterns like this in training

When It Doesn't

Regime break or transition

- Unseen systemic shock — sector stress overrides TD-specific signal (e.g. SVB)
- Phase transition (crisis → recovery) — crisis signals persist into recovery (e.g. AML rebound)
- Signal semantics drift — same feature means different things in different phases
- Ensemble breakdown — at least two models disagree on current regime

Limitations & Next Steps

Limitations

- 01

Thin training data
~800 rows per fold from a single stock. One-off events like regulatory shocks or leadership changes may appear once or not at all.
- 02

Direction only, no magnitude
Predicts which way, not how far. Position sizing needs a separate layer.
- 03

Signals break when the regime changes
The model cannot detect when its learned patterns have stopped applying. Fold 1: SVB panic was never in the training window. Fold 8: AML fine was already priced in but bearish signals kept firing.
- 04

Narrow news coverage
Only TD's own press releases. Broader macro and market news is excluded.

What Would Unlock More

- 01

Panel training across Canadian bank peers
Train on RY, BMO, BNS, CM, NA alongside TD — more observations per period, more crisis and recovery cycles.

→ thin training data
- 02

Magnitude layer
Panel data unlocks more complex features; add options-implied volatility and analyst revision signals.

→ direction only
- 03

Broader news coverage
Objective third-party sources replace TD-only press releases; peer bank language during TD's crisis adds outside perspective.

→ narrow news coverage + signals breaking when regime changes

Section VII • Executive Summary

What We Built



3K+

passages annotated
automatically across 5 years



TD vs XFN

predict TD outperformance vs sector
ETF over 5 trading days



3 + 3

price, language, news read together via
XGBoost ensemble



56.6%

directional accuracy
95% CI [53.1%, 60.1%]



+6.4pp / +5.5pp

over majority class / momentum
baselines



Stable-regime ready

baseline performs reliably in normal
market conditions, with edge cases
flagged

A statistically validated ML system — proven on TD, ready to extend.

APPENDIX

12

SUPPLEMENTAL MATERIAL

Appendix

TD LANGUAGE SIGNALS · PREDICTIVE MODEL

APPENDIX

Appendix

NLP Validation & Topic Taxonomy

VALIDATION

1 · Manual Spot-Check

Stratified sample of 30 passages reviewed against LLM labels and key quotes.

2 · Pre-trained Model Comparison — FinBERT

FinBERT — a pre-trained NLP model fine-tuned on financial text for sentiment classification.

Run independently on 856 transcript passages. **86.8% sign agreement** with LLM scores (Pearson r = 0.501). FinBERT is systematically more cautious — divergence concentrated on passages where the LLM detects optimistic management framing that a lexical model reads as factually neutral.

3 · Known Event Cross-Checks

EVENT	EXPECTED SIGNAL	OBSERVED
AML consent order Oct 2024	regulatory_AML share spike; CEO sentiment drop	✓
BoC rate hiking cycle Mar 2022	macro_outlook share spike; NIM tone improvement	✓
First Horizon collapse May 2023	M_and_A share spike; negative sentiment	✓
Post-AML resolution FY2025Q1+	CEO sentiment recovery toward positive	✓

TOPIC TAXONOMY — 12 LABELS, 1-3 PER PASSAGE

Business Performance 8 labels

NIM	Net interest margin, deposit pricing, spread commentary
credit_quality	Loan losses, provisions, impairments
capital	Capital ratios, dividends, share buybacks
US_retail	TD Bank America segment
Canadian_personal	Canadian Personal & Commercial Banking
wealth_wholesale	TD Wealth + TD Securities
cost_efficiency	Operating expenses, headcount
M_and_A	Acquisitions and divestitures (First Horizon, Schwab stake)

Forward Signals 4 labels

regulatory_AML	AML consent order, compliance, regulatory actions
macro_outlook	Interest rate outlook, inflation, recession risk
guidance	Forward-looking statements, management targets
other	Everything else

Appendix

Feature Engineering

Price & transcript: **trading days** · News: **calendar days**

PRICE MODEL · 6 FEATURES

FEATURE	FORMULA	WHAT IT CAPTURES
td_return_5d	$TD[t]/TD[t-5]-1$	5-day return (~1 wk). Short-term momentum.
td_return_20d	$TD[t]/TD[t-20]-1$	20-day return (~1 mo). Medium-term momentum.
td_vs_xfn_5d	$td_ret5d - xfn_ret5d$	TD vs sector, 5-day window.
td_vs_xfn_20d	$td_ret20d - xfn_ret20d$	TD vs sector, 20-day window.
td_dist_52w_high_v2	$(TD[t]-max[t-251..t]) / TD[t]$	Distance from 52-wk high. ≤ 0 ; near zero = near peak.
td_volatility_20d	$std(daily_ret[t-19..t])$	Rolling 20-day return volatility.

All features use adj_close — corrects for TD's quarterly and XFN's monthly dividends.

TRANSCRIPT MODEL · 10 FEATURES

FEATURE	WHAT IT CAPTURES
exec_tone_ffill	Mean CEO + CFO prepared-remarks sentiment
transcript_analyst_qa_sentiment_mean_ffill	Avg sentiment across analyst Q&A
framing_gap_ffill	CEO verbal tone minus written filing tone
topic_guidance_share_ffill	Share of call discussing forward guidance
topic_guidance_sentiment_ffill	Tone of guidance discussion
topic_regulatory_AML_share_ffill	Share of call discussing AML/regulatory matters
topic_regulatory_AML_sentiment_ffill	Tone of AML/regulatory discussion
topic_credit_quality_share_ffill	Share of call discussing credit quality
topic_credit_quality_sentiment_ffill	Tone of credit quality discussion
days_since_call	Trading days since most recent earnings call (staleness)

NEWS MODEL · 5 FEATURES

FEATURE	WHAT IT CAPTURES
news_sent_mean_7d	Mean sentiment, last 7 calendar days
news_sent_mean_30d	Mean sentiment, last 30 calendar days
news_count_7d	Article count, last 7 days. High count signals negative event clusters.
news_count_30d	Article count, last 30 days. Independent of tone.
days_since_last_news	Calendar days since most recent article — flags coverage gaps.

TRANSCRIPT FEATURE ACTIVATION WINDOW (ILLUSTRATIVE)

QUARTER	CALL DATE	FEATURES ACTIVE UNTIL	TRADING DAYS ACTIVE
FY2021Q1	2021-02-25	2021-05-26	63
...			
FY2025Q3	2025-08-28	2025-12-03	68
FY2025Q4	2025-12-04	2026-04-02	82
FY2026Q1	2026-04-03	2026-04-09 (end)	4

Appendix

Design Choices & Hyperparameters

WHAT WE PICKED

XGBoost vs alternatives

Native multiclass probability output (`multi:softprob`), built-in L1/L2 regularisation, and native missing-value handling — important with ~800 training rows per fold and no separate imputation step.

ALGORITHM	REASON NOT CHOSEN
XGBoost ✓	Selected — native multiclass, regularisation, handles missing values
LightGBM	Tested; no meaningful performance difference on this dataset
Logistic regression	Cannot capture non-linear interactions between features
Neural networks	Inappropriate for this data scale (~800 rows per fold)

Three separate models vs single combined model

Transcript features forward-fill for 56–82 trading days; mixing near-static signals with daily price and news in one model produces misleading feature interactions. Separate models also allow each signal source to be inspected and fixed independently.

Hard majority vote vs soft-prob averaging

Achieved 56.6% directional accuracy vs. 53.6% for soft-prob averaging. The models predict direction reliably but are not well-calibrated in probability space — hard voting uses the directional signal and ignores the miscalibrated confidence.

WHAT WE TESTED AND REJECTED

FEATURE / CHOICE	REASON REJECTED
60-day price returns	Adds noise at 5-day horizon; quarterly retraining already captures regime shifts
Volume change features	No directional mechanism; empirically reduced performance
90-day news window	Dilutes short-horizon signal
<code>call_decay</code> on NLP features	Near-perfect multicollinearity; replaced by <code>days_since_call</code>
Macro features (FX, CPI, rates)	Already embedded in XFN benchmark; would double-count sector movement
Single combined model	Loss of interpretability; misleading feature interactions between stale and fresh signals
Soft-prob averaging	Underperformed hard vote; models are not probability-calibrated

HYPERPARAMETER SEARCH

PARAMETER	SEARCH RANGE
<code>max_depth</code>	2, 3, 4
<code>min_child_weight</code>	10, 20, 30
<code>n_estimators</code>	50, 80, 120
<code>reg_alpha</code> (L1)	0.0, 0.2
<code>reg_lambda</code> (L2)	1.0, 2.0
108 configs x 3 models · Criterion: $0.6 \times \text{dir_accuracy} + 0.4 \times \text{macro_F1}$	

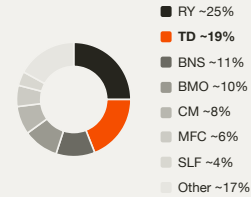
BEST CONFIGURATIONS

MODEL	DEPTH	MIN_CHILD	N_EST	A	A
Price	3	30	120	0.0	1.0
Transcript	4	20	80	0.2	1.0
News	2	30	50	0.2	2.0

Best parameters tuned 2026-04-23 on folds 4–10.

XFN BENCHMARK COMPOSITION

XFN (iShares S&P/TSX Capped Financials ETF) is TD's most direct liquid peer benchmark, with the major Canadian banks as its top holdings.



Approximate weights, rebalanced periodically. Source: iShares Canada.

Appendix

Evaluation Methodology & Per-Fold Performance

EVALUATION METHODOLOGY

Stride-Offset Averaging

The 5-day forward return target means consecutive test days share 4 days of return data — they are not independent. To avoid inflating estimates, every 5th observation is taken (a non-overlapping subset). Metrics are computed for each of the 5 valid offsets (0–4) and averaged. All test days are used across the 5 offsets combined.

Metric Definitions

METRIC	DEFINITION
directional_call_rate	Share of days where prediction ≠ 0
directional_accuracy	Share of directional calls where sign(prediction) = sign(actual); random baseline = 0.50
auc_pos	OVR AUC for +1 class using softmax P(+1); diagnostic only
auc_neg	OVR AUC for −1 class using softmax P(−1); diagnostic only

Why These Metrics

Mean accuracy is excluded because it is dominated by the neutral class — a model that always predicts neutral would score well without any genuine directional signal. `directional_accuracy` measures only the calls the model commits to. AUC is retained as a diagnostic on sub-model probabilities only and does not factor into ensemble selection.

HARD MAJORITY VOTE — PER-FOLD PERFORMANCE

FOLD	TRAIN PERIOD	TEST PERIOD	DIR. CALL RATE	DIR. ACCURACY
1	2021-02-25 → 2023-02-28	2023-03-08 → 2023-06-06	0.968	0.361
2	2021-05-27 → 2023-05-30	2023-06-07 → 2023-09-06	0.951	0.694
3	2021-08-26 → 2023-08-29	2023-09-07 → 2023-12-05	1.000	0.538
4	2021-11-25 → 2023-11-28	2023-12-06 → 2024-03-07	0.968	0.657
5	2022-02-28 → 2024-02-29	2024-03-08 → 2024-06-06	1.000	0.572
6	2022-05-30 → 2024-05-30	2024-06-07 → 2024-09-06	0.936	0.594
7	2022-08-29 → 2024-08-29	2024-09-09 → 2024-12-05	1.000	0.686
8	2022-11-28 → 2024-11-28	2024-12-06 → 2025-03-10	0.985	0.388
9	2023-03-01 → 2025-03-03	2025-03-11 → 2025-06-09	1.000	0.459
10	2023-05-31 → 2025-06-02	2025-06-10 → 2025-09-09	1.000	0.663
11	2023-08-30 → 2025-09-02	2025-09-10 → 2025-12-08	0.968	0.457
12	2023-11-29 → 2025-12-01	2025-12-09 → 2026-03-11	0.985	0.691
13	2024-03-01 → 2026-03-04	2026-03-12 → 2026-04-09 20d – partial	1.000	0.600